Daily Practice Problems (DPP) - Power BI

Easy Level

1. **Describe the steps to download and install Power BI Desktop.**

Answer.

- Search "Power BI Desktop download" on Google.
- Open the official Microsoft website and Click on the download button.
- Run the setup file downloading.
- Complete the installation and open Power BI Desktop

2. **How do you import an Excel file into Power BI**

Answer.

- Open Power BI Desktop and Go to the tab.
- Click on get data and select Excel Choose the Excel file and click load to import the data.

3. **What is the purpose of Power Query in Power BI**

Answer.

- Power Query is used for data cleaning and transformation. It helps to:
- Remove and replace missing values Filter and sort.
- Split and merge columns Change data types.

4. **Explain the difference between Report View, Data View, and Model View in Power BI.**

Answer.

Report View: Use to create chart and visual reports.

Data View: Used to see data in table format.

Model View: Used to manage relationship between tables

Medium Level.

5. **How can you connect Power BI to an SQL Server database? List the steps involved.**

Answer.

- Open Power BI Desktop and click home and get data SQL Server.
- Enter the server name and database name.
- Choose import file or DirectQuery mode.
- Click load and transform data.
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6. **Describe how to handle missing values in Power Query**

Answer.

Click Transform data to open power query editor and select the column with missing value.

Use replace values to replace null value and use remove rows to delete rows with missing value.

Use Fill up or fill down if required.

7. **What is the significance of relationships between tables in Power BI? How do you create them**

Answer.

Relationships connect table using a common column. They ensure accurate filtering and calculation across multiple tables.

To create relationships:

- Go to Model View and drag a column from one table to the related column in another table.
- Define the relationship type (one-to-many).

8. Explain the process of publishing a Power BI report to the Power BI service.

Answer.

Complete the report in Power BI Desktop and click on the home tab.
Select publish and choose a workspace. The report will be uploaded to Power BI service.

Hard Level

9. Explain the difference between Merge Queries and Append Queries in Power Query. Provide a use case for each.

Answer:

Merge Queries: Combines two tables based on a common column (like a join).

Use case: Combine customer and Orders tables using Customer ID.

Append Queries: Adds one table below another table (like stacking rows).

Use case: Combining sales data of 2022 and 2023.

10. How would you optimize a Power BI report that is running slow due to large datasets

Answer:

Remove unnecessary columns and use measures instead of calculated columns.
Reduce the number of visuals and use import mode when possible.
Optimize the data model.

11. Discuss the different data sources that Power BI supports and the advantages of using DirectQuery over import mode.

Answer: Excel, SQL, Web, SharePoint, Azure

Advantages of DirectQuery:

- Provides real-time data.
- Handles large datasets and No need to store data inside Power BI.

12. Describe a real-world scenario where Power BI can be used to enhance data-driven decision-making.

Answer: A retail company can use Power BI to analyze sales performance, track profit by region, and understand customer behavior. This helps management make better decisions regarding pricing, marketing, and inventory management